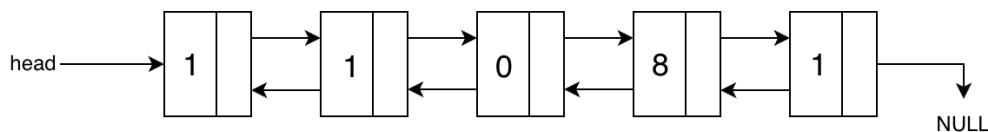


Task Description | End-Sem Practical Examination | EE-A | Batch 1

Write a C program that accepts the **last 5 digits** of your own 11-digit roll number as an array of digits. The program should perform sorting, array reversal, and Doubly Linked List operations based on the roll number. For example, if the roll number is **25295918011**, then, the last five digits are **18011** and it should be stored as {1, 8, 0, 1, 1}.

Tasks to be performed

1. Input the last 5 digits of the roll number into an integer array.
2. Choose and apply sorting algorithm (non-decreasing order)
 - a. Preserve the original copy of the array for later.
 - b. Convert the digit array into a complete 5-digit roll number and compute:
roll_number % 3
 - i. if $\text{roll_number} \% 3 == 0$, use Merge Sort
 - ii. if $\text{roll_number} \% 3 == 1$, use Quick Sort
 - iii. if $\text{roll_number} \% 3 == 2$, use Heap Sort
3. Print the sorted 5-digit roll number. For example, **18011** will be printed as **01118** (without spaces)
4. Reverse the original copy of the roll number in the array itself. For example, {1, 8, 0, 1, 1} will be reversed as {1, 1, 0, 8, 1}.
5. Print the reversed array. For example, {1, 1, 0, 8, 1} will be printed as **11081**.
6. Create a Doubly Linked List using the reversed roll number in the reversed order (Example shown below).



7. Traverse the LL in forward direction and print the reversed order. For the above example, the output would be **11081**.
8. Traverse the LL in the backward direction and print the original order. For the above example, the output would be **18011**.

Important: Each print output must be on a new line. If the given example outputs do not contain whitespaces between digits, then your output must also not contain whitespaces.